SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

Course Code:	CSA10 / CSA1063	**Course Title:**	Software Engineering
CO Assessed:	CO2	**Assessment:**	Assessment Tool 2 – Scenario-Based Requirement Analysis
Weightage:	20%	**Total Marks:**	25
CO2 Statement:	*Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills*		
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Date:	4.08.2026	**Duration:**	60 minutes

Code of Conduct
I ________Rashmi Naurene (192521357)___________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the Student: ______Rashmi Naurene_______

Annexure A – Scenario-Based Requirement Analysis
Instructions to Students:
Each student will be assigned an individual software project problem statement. Analyze the given scenario and prepare a complete Software Requirement Analysis document by following the steps below.
Question
Based on the **assigned problem statement**, perform a **Scenario-Based Requirement Analysis** and prepare a Software Requirement Specification (SRS) document. Your analysis should include the following:

1. **Study and Analyze the Scenario**
   - Carefully understand the given problem statement.
   - Identify the business domain, stakeholders, existing challenges, and system objectives.
2. **Identify Functional and Non-Functional Requirements**
   - List all functional requirements required by the proposed system.
   - Identify suitable non-functional requirements such as performance, security, reliability, usability, scalability, maintainability, and availability.
3. **Identify Actors and Use Cases**
   - Identify all primary and secondary actors interacting with the system.
   - List the major use cases associated with each actor.
   - Draw a Use Case Diagram representing the interactions between actors and the system.
4. **Prepare the Software Requirement Specification (SRS)**
   Develop an SRS document containing:
   - Introduction
   - Problem Description
   - Scope of the System
   - Functional Requirements
   - Non-Functional Requirements
   - Use Case Descriptions
   - Data Requirements
   - External Interface Requirements
   - System Features
5. **Identify Assumptions and Constraints**
   - List the assumptions considered while developing the system.
   - Identify technical, operational, economic, legal, and time-related constraints affecting the project.

6. **Validate Requirement Completeness and Consistency**
    - Verify whether all stakeholder requirements have been addressed.
    - Check for ambiguity, redundancy, inconsistency, and missing requirements.
    - Suggest improvements to enhance the quality and completeness of the requirement specification.

1 · Scenario Understanding & Problem Analysis

1.1 Business Domain & Context

The assigned scenario sits within the Human Resources and Learning & Development (L&D) domain, specifically the sub-area of workforce capability management. Organizations across fast-moving industries such as IT services, manufacturing, and finance are finding that the skills their employees hold gradually drift out of alignment with the skills their roles actually demand. This is not a hypothetical risk — new tools, frameworks, and compliance requirements emerge continuously, and traditional annual appraisal cycles are too infrequent to catch the gap early. The domain therefore calls for a system that treats skill measurement as a continuous, data-driven activity rather than a once-a-year paperwork exercise.

1.2 Stakeholders

Four groups of people and one automated component have a direct stake in the system's behaviour, each with a different relationship to the data it produces.

- Employees, who want an honest, current picture of their own competencies and a clear next step to close any gap.
- Reporting Managers, who need to see gaps across their whole team at a glance so coaching and resourcing decisions are based on evidence rather than guesswork.
- HR / L&D teams, who own the competency frameworks and the training catalogue, and who are ultimately accountable for workforce readiness metrics reported to leadership.
- Senior Leadership, who consume aggregated analytics to plan budgets and long-term capability investment rather than interacting with individual records.
- The Recommendation Engine, an internal system actor that automatically compares profiles against frameworks and ranks training suggestions — it is not a human stakeholder, but its correctness is central to every other stakeholder's trust in the system.

1.3 Existing Challenges

Three recurring problems motivate the project. First, performance evaluations are typically qualitative and manager-dependent, which makes it hard to compare skill levels consistently across a department. Second, training assignment today is largely generic — employees are pointed at broad course catalogues rather than courses targeted at their specific deficiency, which wastes time and lowers completion rates. Third, HR has no simple way to see workforce-wide skill coverage, so decisions about hiring versus upskilling are made without a clear evidence base.

1.4 System Objectivesj

- Continuously evaluate employee competencies from performance reviews, certifications, assessments, and project experience.
- Identify skill deficiencies automatically by comparing profiles against role-specific competency frameworks.
- Recommend personalized, ranked training programs that target the specific gap identified.
- Give managers and HR dashboards and reports that summarize readiness at the team and organizational level.
- Track learning progress and certification completion so gaps are demonstrably closing over time.
- Support scalable, data-driven decisions in training investment and career development planning.

2 · Software Requirement Specification (SRS)

2.1 Introduction

This document specifies the software requirements for the Employee Skill-Gap Analysis and Training Recommendation System. It is intended for use by the development team, the HR/L&D stakeholders who will operate the platform, and the course evaluator assessing this submission. The specification follows a standard SRS structure covering purpose, scope, functional and non-functional requirements, actors and use cases, data and interface requirements, and system features.

2.2 Problem Description

Organizations need a platform that evaluates employee competencies against predefined job-role requirements, using data collected from performance reviews, certifications, assessments, and project experience. The platform must identify skill deficiencies through data analytics and recommend personalized training to close them. Managers require dashboards to monitor progress, the system must support role-based learning paths and competency frameworks per department, and automated notifications should drive completion of assigned learning activities.

2.3 Scope of the System

The system covers profile data collection, automated gap analysis against configurable competency frameworks, ranked training recommendations, manager and HR reporting dashboards, learning-path sequencing, and notification delivery. It does not include payroll processing, recruitment/applicant tracking, or the authoring of training content itself — the platform consumes an existing or externally supplied course catalogue rather than producing one. Integration with the organization's HRMS for identity and existing performance-review data is in scope; building a replacement HRMS is not.

2.4 Functional Requirements

Functional requirements describe what the system must do, grouped by capability area.

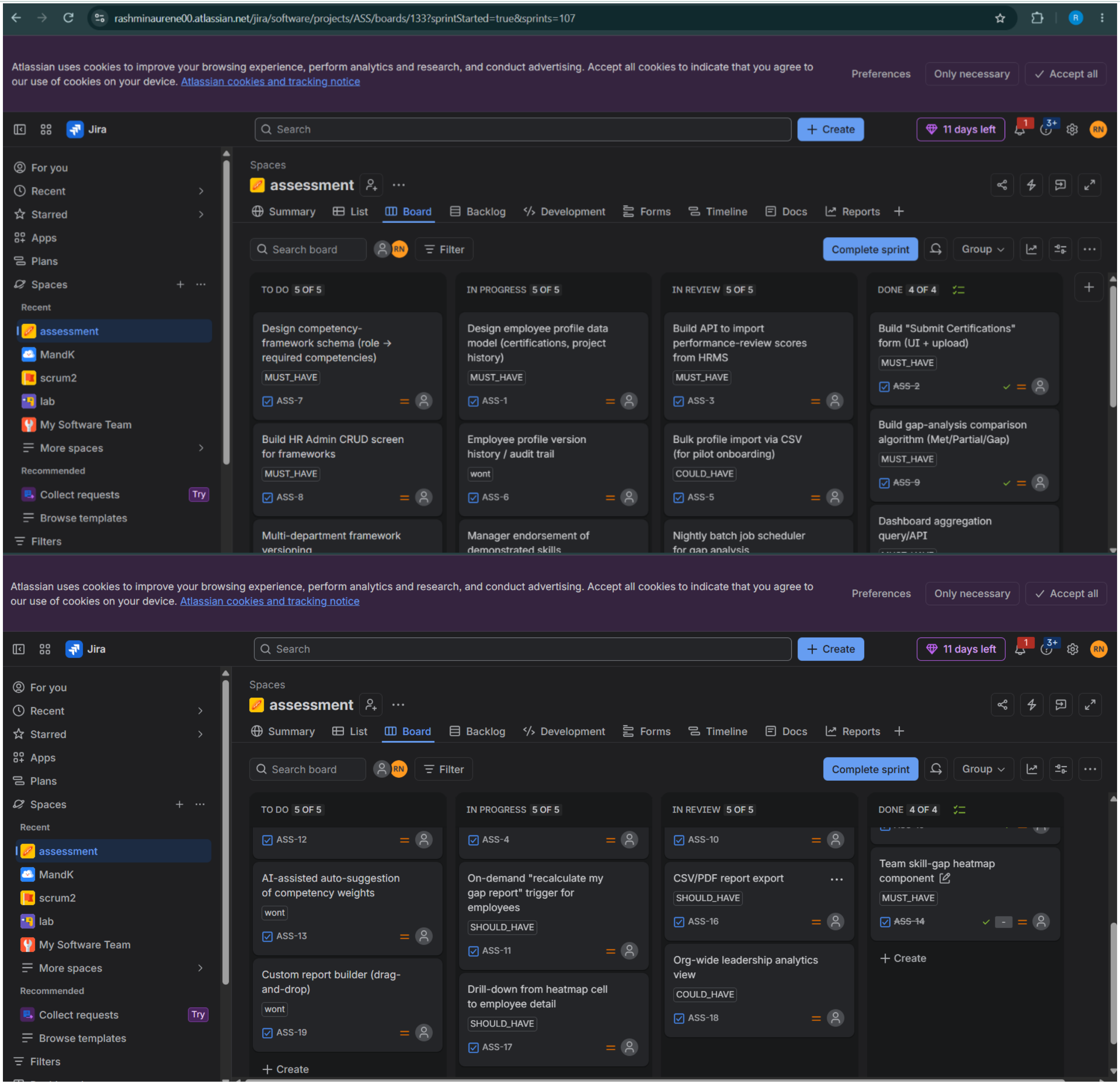
ID	Requirement
FR-01	The system shall allow an employee to create and update a competency profile including certifications and project experience.
FR-02	The system shall import performance-review scores from the HRMS on a scheduled basis.
FR-03	The system shall allow HR administrators to define and edit competency frameworks per job role and department.
FR-04	The system shall compare an employee's profile against the applicable competency framework and produce a Met / Partial / Gap classification for each competency.
FR-05	The system shall generate a ranked list of training recommendations for each identified gap, based on expected competency impact.
FR-06	The system shall allow employees to accept, defer, or decline a recommended training item.
FR-07	The system shall sequence accepted training into a role-based learning path.
FR-08	The system shall send automated reminders for assigned but incomplete training as the due date approaches.
FR-09	The system shall provide managers with a team-level skill-gap heatmap dashboard.

FR-10	The system shall allow HR to export competency and training-completion reports for a selected department or date range.
FR-11	The system shall allow managers to approve or reject training budget requests linked to a recommendation.
FR-12	The system shall restrict access to competency data and dashboards according to role-based permissions.

2.5 Non-Functional Requirements

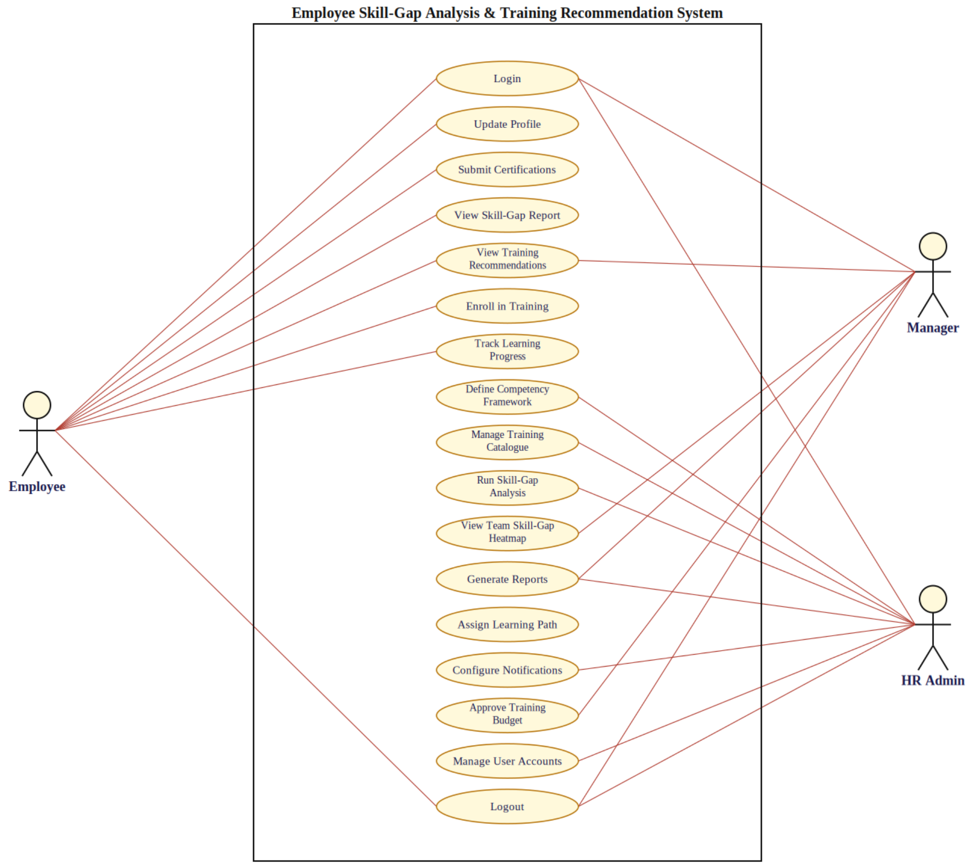
Category	Requirement
Performance	The gap-analysis batch job shall process at least 5,000 employee profiles within a 30-minute nightly window.
Security	All competency and performance data shall be encrypted in transit (TLS 1.2+) and at rest, with role-based access control enforced on every API endpoint.
Reliability	The platform shall maintain 99.5% uptime during business hours, excluding scheduled maintenance windows.
Usability	A first-time employee user shall be able to submit a certification and view their skill-gap report without external training, in under 5 minutes.
Scalability	The architecture shall support horizontal scaling to accommodate organizational growth from 1,000 to 50,000 employee profiles without redesign.
Maintainability	Competency frameworks and course-catalogue mappings shall be editable by HR administrators through configuration, without requiring code changes.
Availability	Dashboards and reports shall be accessible 24/7, with scheduled maintenance windows communicated at least 48 hours in advance.
Auditability	Every change to a competency framework or a training recommendation override shall be logged with a timestamp and the responsible user.

The following is activity and requirement prioritizing and organizing in jira software



2.6 Actors and Use Cases

Three primary actors interact directly with the system. The Employee is the main data contributor and consumer of recommendations. The Manager consumes team-level analytics and approves training-related decisions. The HR Admin configures the frameworks and catalogue that the analysis engine depends on, and owns organization-wide reporting. The Use Case Diagram below shows each actor's interactions with the system's core use cases.



Actor Summary

Actor	Role Summary
Employee	Primary actor; maintains their own profile, views their skill-gap report, and acts on training recommendations.
Manager	Primary actor; reviews team-level skill gaps, approves training budgets, and monitors learning-path progress for direct reports.
HR Admin	Primary actor; defines competency frameworks, manages the training catalogue, runs organization-wide reporting, and administers user accounts.

2.7 Use Case Descriptions

Detailed descriptions are provided below for the six use cases central to the system's core value proposition.

UC-01 — Run Skill-Gap Analysis	
Actor(s)	HR Admin (initiates); System (executes)
Precondition	An employee profile is complete and a competency framework exists for their role.
Main Flow	The system compares the employee's recorded competencies against the role's required competency matrix and produces a Met / Partial / Gap classification with a deficiency score for each competency.
Postcondition	A skill-gap report is available to the employee and their manager.
UC-02 — View Training Recommendations	
Actor(s)	Employee

Precondition	At least one competency is classified as Gap for the employee.
Main Flow	The employee opens their dashboard; the system displays the top-ranked courses for each open gap, ordered by expected competency impact.
Postcondition	The employee can accept, defer, or decline each recommendation.
UC-03 — Define Competency Framework	
Actor(s)	HR Admin
Precondition	The HR Admin is authenticated with administrator privileges.
Main Flow	The HR Admin creates or edits the list of required competencies and proficiency levels for a specific job role.
Postcondition	The updated framework is used in all subsequent gap-analysis runs for that role.
UC-04 — View Team Skill-Gap Heatmap	
Actor(s)	Manager
Precondition	The manager has at least one direct report with a completed skill-gap report.
Main Flow	The manager opens the Team Dashboard; the system renders a colour-coded matrix of competency versus team member.
Postcondition	The manager can drill into any cell to see the underlying employee-level detail.
UC-05 — Enroll in Training	
Actor(s)	Employee
Precondition	A training recommendation has been accepted by the employee.
Main Flow	The employee confirms enrolment; the system reserves a seat (where applicable) and adds the item to the employee's active learning path.
Postcondition	The training item's status changes to In Progress and a reminder schedule is created.
UC-06 — Generate Reports	
Actor(s)	HR Admin, Manager
Precondition	The requesting user has reporting permissions for the selected scope.
Main Flow	The user selects a department and date range; the system aggregates competency and completion data into an exportable report.
Postcondition	A report file is generated and made available for download.

2.8 Data Requirements

The system must persist and manage the following core data entities.

- Employee Profile — identity reference, role, department, certifications, project history.
- Competency Framework — role, competency name, required proficiency level, department applicability.
- Assessment Record — source (review, certification, project), competency, measured proficiency, date.
- Skill-Gap Report — employee, competency, classification (Met/Partial/Gap), deficiency score, generated date.
- Training Catalogue Entry — course name, provider, duration, mapped competencies, format (online/in-person).
- Recommendation — employee, competency gap, ranked course list, status (pending/accepted/deferred/declined).

- Learning Path — employee, ordered list of training items, status, target completion date.
- Notification Log — recipient, channel, message, sent timestamp, read status.

2.9 External Interface Requirements

Interface	Description
HRMS Integration	REST API connection to the organization's HRMS for employee identity, role, and performance-review score import.
Single Sign-On (SSO)	SAML 2.0 / OAuth 2.0 integration so employees authenticate using existing corporate credentials.
Notification Channels	Email (SMTP) and in-portal notification delivery for training reminders and recommendation alerts.
Reporting Export	CSV and PDF export interfaces for HR and manager reports.
Training Catalogue Feed	Optional API/CSV import interface for onboarding external training-provider catalogues.
Web User Interface	Responsive browser-based UI accessible on desktop and tablet, supporting current major browsers.

2.10 System Features

- Competency Profile Builder — guided form for employees to submit certifications and project history.
- Automated Gap-Analysis Engine — nightly batch and on-demand comparison of profiles against frameworks.
- Recommendation Ranking Engine — impact-based ordering of catalogue courses against open gaps.
- Manager & HR Dashboards — heatmaps, completion tracking, and exportable reports.
- Role-Based Learning Paths — sequenced training aligned to each employee's role and gaps.
- Notification Scheduler — configurable reminders across email and in-portal channels.
- Framework & Catalogue Administration — HR-facing configuration screens requiring no code changes.

3 · Assumptions & Constraints

3.1 Assumptions

- All employees already hold a valid HRMS account that can be linked via SSO.
- Job-role competency frameworks can be defined by HR without requiring input from external subject-matter consultants for the pilot rollout.
- The initial training catalogue is curated and reasonably complete for the pilot departments before go-live.
- Performance-review data is exported from the HRMS in a consistent, machine-readable format.
- Employees have basic digital literacy sufficient to use a standard web application without dedicated training.

3.2 Constraints

Type	Constraint

Technical	The system must integrate with the organization's existing HRMS API, whose data schema cannot be modified by this project.
Operational	Competency frameworks must be reviewed and approved by HR before each rollout to a new department.
Economic	The pilot budget limits the initial training-catalogue integration to a curated subset of course providers.
Legal	Employee competency and performance data handling must comply with applicable data-protection regulations (e.g., consent, retention limits).
Time	The pilot must be demonstrable within a single sprint cycle, limiting Sprint 1 scope to the core profile-to-recommendation flow.

4 · Requirement Validation

4.1 Stakeholder Requirement Coverage

Each stakeholder group identified in Section 1.2 maps to at least one functional requirement and one use case: Employees are covered by FR-01, FR-06, FR-07 and UC-02/UC-05; Managers by FR-09, FR-11 and UC-04; HR Admins by FR-03, FR-10, FR-12 and UC-03/UC-06; and the Recommendation Engine by FR-04 and FR-05. No stakeholder identified in the scenario analysis is left without a corresponding requirement, which was verified by cross-checking Section 1.2 against the functional requirement table in Section 2.4.

4.2 Ambiguity, Redundancy, and Inconsistency Check

The requirement set was reviewed for three common defect types. No redundant requirements were found — each FR addresses a distinct capability. One ambiguity was identified and resolved during drafting: the original scenario phrase "identify skill deficiencies using data analytics" was under-specified, so it was translated into the concrete, testable FR-04 with an explicit Met/Partial/Gap classification rule. No direct inconsistencies were found between functional and non-functional requirements, though FR-02 (scheduled HRMS import) and the Performance NFR (30-minute nightly batch window) were cross-checked to confirm the import step fits inside the analysis window rather than competing with it.

4.3 Suggested Improvements

- Add an explicit data-retention NFR specifying how long assessment history is kept, to fully satisfy the Legal constraint in Section 3.2.
- Extend FR-06 with a defined SLA for how long a deferred recommendation remains valid before it is re-ranked.
- Introduce a lightweight feedback loop use case allowing employees to rate completed training, which would improve the accuracy of the Recommendation Ranking Engine over time.
- Clarify in a future revision whether external Training Content Providers (Section 1.2) require a separate authenticated interface, since the current scope treats the catalogue as a one-way import only.

Rubrics for Calculation

Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Scenario Understanding and Problem Analysis (30%)	Demonstrates thorough understanding of the scenario, stakeholders, objectives, and business context with clear analysis.	Demonstrates good understanding with minor omissions.	Basic understanding with limited analysis.	Poor understanding; major aspects missing or incorrect.
Functional and Non-Functional Requirement Identification (25%)	Clearly identifies comprehensive, accurate, and well-classified functional and non-functional requirements.	Identifies most requirements with minor classification errors.	Identifies only basic requirements; several omissions.	Requirements are incomplete, inaccurate, or poorly classified.
Actor Identification and Use Case Modelling (20%)	Correctly identifies all relevant actors and use cases; use case diagram is complete, accurate, and follows UML standards.	Most actors and use cases identified; diagram has minor errors.	Limited actors/use cases identified; diagram partially correct.	Actors/use cases missing or incorrect; diagram absent or inaccurate.
Software Requirement Specification (SRS) Documentation (15%)	SRS is well-structured, complete, professionally organized, and includes all required sections.	SRS is organized with minor missing sections or formatting issues.	SRS includes only essential sections with limited detail.	SRS is incomplete, poorly organized, or lacks essential components.
Assumptions, Constraints, and Requirement Validation (10%)	Clearly identifies realistic assumptions and constraints; validates requirements for completeness, consistency, and ambiguity with appropriate justification.	Identifies most assumptions and constraints; validation is adequate.	Limited assumptions/constraints; validation is superficial.	Assumptions, constraints, and validation are missing or incorrect.

Criteria	**Mark**
Scenario Understanding and Problem Analysis (30%)	/5
Functional and Non-Functional Requirement Identification (25%)	/5
Actor Identification and Use Case Modelling (20%)	/5
Software Requirement Specification (SRS) Documentation (15%)	/5
Assumptions, Constraints, and Requirement Validation (10%)	/5
TOTAL	/25